

# MAZHARUL ISLAM

Software Engineer

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## SUMMARY

Building products and sharing what I learn along the way. Backend, frontend, mobile, infrastructure, DevOps, ML - all of it, because agents made the parts I used to hand off cheap enough to keep. One person can hold a whole product now. Still working out what that changes, which is turning out to be most of the job.

## TECHNICAL SKILLS

**Languages and Frameworks:** Python • Django • Django REST Framework • FastAPI • Dart • Flutter • TypeScript • Next.js • React

**Databases and Messaging:** PostgreSQL • Redis • Celery • Celery Beat

**AI and NLP:** Speech Recognition • NLP Processing • LLM Integration • PyTorch • WhisperX • OpenAI API

**Infrastructure and DevOps:** Docker • Nginx • GCP • DigitalOcean • CI/CD • Prometheus • Grafana • Firebase • FCM

**AI-Assisted Development:** Claude Code • Cursor • Agentic Workflows • AI Pair Programming

## PROFESSIONAL EXPERIENCE

**LII Lab** Sylhet, Bangladesh

**Senior Software Engineer** 01/2025 - 04/2026

*Led backend engineering for OneIELTS and OnePTE, AI-powered English test preparation platforms.*

- Architected a multi-tenant SaaS backend with django-multitenant: full data isolation, custom domain routing, per-tenant scoring config and feature flags.
- Designed a QTI 3.0-compliant exam engine supporting 100+ question types across IELTS Academic and General, with XML parsing, interaction routing, and strategy-pattern band-score normalization.
- Built a real-time speech evaluation pipeline integrating speech recognition, pronunciation assessment and NLP processing, with end-to-end scoring in under 15 seconds.
- Engineered multi-gateway payment infrastructure across 5 providers, with subscription lifecycle management and webhook idempotency.
- Built the Studio API layer for the moderator platform: content authoring, expert evaluation queue, exam rejudge pipelines, and KPI analytics.
- Set up production observability with Prometheus and Grafana: health monitoring, per-API latency instrumentation, and alerting.
- Mentored junior engineers through code reviews, PR feedback, and architectural walkthroughs.

**Stack:** Django • DRF • Python • PostgreSQL • Redis • Celery • FastAPI • Docker • GCP • Prometheus • Grafana • Next.js • TypeScript

Full-stack ownership of OnePTE: designed and shipped the Flutter app from day one and built the Django AI scoring backend.

- Designed and built the Flutter cross-platform app from scratch: all four PTE modules with 20+ task types, audio recording, and timer management.
- Built a task-group-based mock test engine with modular exam templates, configurable time allocations, automated question progression, and multi-dimensional score breakdowns.
- Engineered the AI scoring backend for spoken and written tasks using speech recognition, pronunciation assessment and NLP processing for multi-trait evaluation.
- Integrated subscription billing across 4 platforms with webhook handling, transaction deduplication, and in-app purchase verification.
- Built the Django admin and private API for content moderation, question bank management, subscription analytics with regional reporting, and user acquisition dashboards.
- Led 93+ production releases across Android, iOS and Web; managed release pipelines, Firebase config, and CI/CD across all channels.

Stack: Flutter · Dart · Riverpod · Django · DRF · Python · PostgreSQL · Redis · Celery · Firebase · Next.js · TypeScript

PROJECTS

Sushi Lab, Bilingual restaurant ordering and marketing platform, Chartres, France 2026

- Ordering engine with a hydration-safe cart, scheduled-ahead time slots, per-item option pickers, and server-enforced availability windows
- Triple-channel notifications for every order, reservation and contact submission, dispatched in parallel with strict cross-channel data parity
- AI-discoverable SEO: JSON-LD, dynamic sitemap, dynamic OG images, per-locale canonical and hreflang, and llms.txt
- GA4 ecommerce funnel plus reservation, contact, language-switch and Web Vitals events

Stack: Next.js 16 · React 19 · TypeScript · Tailwind · Zustand · React Hook Form · Zod · React Email · next-intl · Telegram Bot API · GA4

Projecto, University course and proposal management system 2022

- Three roles with distinct permissions: student, supervisor and admin
- PDF and CSV export for project reports
- Background job service for scheduled operations

Stack: Django · Python · PostgreSQL · AJAX · jQuery · Bootstrap

TestForge, Mutation-driven test-backfill agent 2026

- Runs mutmut, reads the surviving mutants, and writes targeted tests to kill them
- Coverage is a proxy; mutation score is the thing coverage was trying to measure

Stack: Python · Claude API · mutmut · pytest

COMPETITIVE PROGRAMMING AND ACHIEVEMENTS

|                                                                                                 |                                                                                                                             |                                                                                                 |
|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 2000+ problems solved · 300+ contests · 1603 cf rating · 1965 cc rating                         | <b>Champion</b> , Champion at the National Hackathon, LU CSE Carnival                                                       | <b>Outstanding Leadership Volunteer Award</b> , IEEE Outstanding Leadership Volunteer Award     |
| <b>Runner-up, Bangladesh</b> , Runner-up in Bangladesh at IEEEExtreme 16.0                      | <b>Champion</b> , Champion at LU TechStorm 4                                                                                | <b>51st, Dhaka Regional Main Round</b> , Ranked 51st at the ICPC Dhaka Regional 2020 Main Round |
| <b>Honorable Mention</b> , Honorable Mention at the ICPC Asia Dhaka Regional Online Preliminary | <u>Codeforces</u> (Max 1603) · <u>CodeChef</u> (Max 1965) · <u>LightOJ</u> · <u>Toph</u> · <u>StopStalk</u> · <u>Vjudge</u> |                                                                                                 |

EDUCATION

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Leading University

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering · GPA 3.6 / 4.0

Sep 2018 - Dec 2022

VOLUNTEERING

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IEEE Computer Society, LU Student Branch Chapter

Chair

Apr 2022 - May 2023

- Co-founded the chapter and launched its first community initiatives
- Organized seminars, webinars and technical events with industry speakers
- Grew membership by more than 20% through outreach programmes
- Delivered a multi-session Flutter bootcamp
- Served as judge and problem setter for programming contests

Leading University Computer Club

ACM Coordinator

Jan 2022 - Dec 2023

- Conducted peer-to-peer sessions on data structures and algorithms
- Mentored juniors in competitive programming across Codeforces, CodeChef and LightOJ
- Organized regular contest practice to prepare members for ICPC and national competitions

Leading University Sports Club

Joint General Secretary

2022 - 2023

- Elected to the executive committee
- Organized inter-department tournaments and university sporting events